Darius Muglich

Machine learning researcher · DPhil student, University of Oxford

dariusmuglich.com / Google Scholar

Research in reinforcement learning, multi-agent systems, generalization, and optimization.

Programming & frameworks: Python, C++, PyTorch, JAX.

Education

University of Oxford 2023–present · Expected September 2027

DPhil · Autonomous Intelligent Machines and Systems (AIMS CDT)

Supervisor: Jakob Foerster

Thesis on policy selection in reinforcement learning: how learning biases and structural constraints shape learned behaviour and generalization.

Current work on logic gate networks and neurosymbolic robot control.

University of Oxford 2020–2021

MSc · Advanced Computer Science

University of British Columbia 2015–2020

BSc · Combined Honours in Mathematics and Computer Science

Selected publications

Ad-Hoc Teamwork from Human Demonstrations NeurIPS 2026

Darius Muglich, Niklas Lauffer, Tin Dizdarevic, Jakob Foerster

Recovered latent human conventions from unlabeled demonstrations to construct diverse evaluation partners, showing that partner coverage can change agent rankings.

Expected Return Symmetries ICLR 2025

Darius Muglich, Johannes Forkel, Elise van der Pol, Jakob Foerster

Learned expected-return-preserving symmetries beyond environment symmetries, enabling symmetry-based coordination without hand-specified transformations.

Equivariant Networks for Zero-Shot Coordination NeurIPS 2022

Darius Muglich, Christian Schroeder de Witt, Elise van der Pol, Shimon Whiteson, Jakob Foerster

Developed equivariant policy architectures and a test-time symmetrization operator to improve zero-shot coordination without retraining existing policies.

Generalized Beliefs for Cooperative AI ICML 2022

Darius Muglich, Luisa Zintgraf, Christian Schroeder de Witt, Shimon Whiteson, Jakob Foerster

Developed autoregressive transformer belief models that generalize to unfamiliar partner conventions and improve ad-hoc teamplay through belief-based search.

Research and industry experience

UC Berkeley · Center for Human-Compatible AI Jun–Sep 2025

Visiting Researcher

Developed a pipeline combining latent-variable modelling, search over human conventions, and KL-regularized PPO to construct playable evaluation partners.

Bazaarvoice Aug 2022–Sep 2026

Data Scientist V · Full-time initially; part-time during the doctorate

Primary engineer for a bandit-based campaign optimization method, delivering an estimated 40% lift in incremental revenue over an analyst-designed heuristic in a multi-month company A/B test.

Selected awards

Toshiba Research and EPSRC CDT Studentship · Fully funded doctoral studentship

NSERC Undergraduate Student Research Award · Three awards

Trek Excellence Scholarship · Top 5% in year, faculty and school at UBC